

Real-Time Data Monitoring of an Industry 4.0 Assembly Line using Pervasive Augmented Reality: First Impressions

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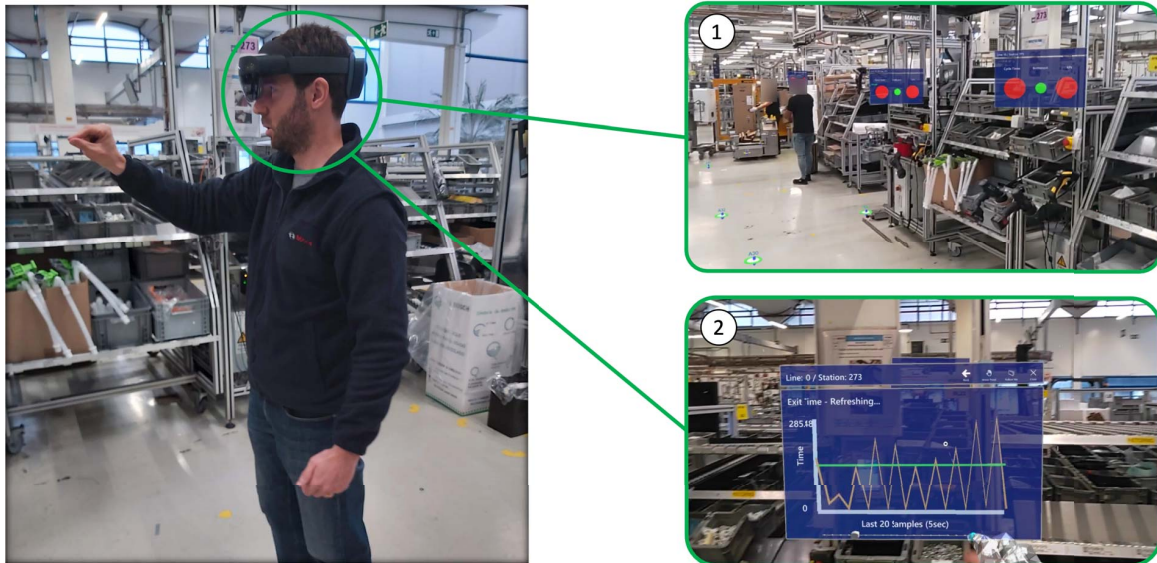


Figure 1: Industrial scenario, having an assembly line manager visualizing real-time data of the whole line (1), as well as detailed information for a given station (2) through Pervasive Augmented Reality (AR) in a Head-Mounted Display (HMD).

ABSTRACT

Industry 4.0 is the latest industrial revolution, tend to seamlessly integrate digital and physical worlds into industrial procedures in an effort to better support different stakeholders' needs. This work proposes a Pervasive Augmented Reality (AR) first prototype, developed in collaboration with industrial partners, to support real-time data monitoring and problem detection of an industrial assembly line. A Human-Centered Design (HCD) methodology was used to identify stakeholders' difficulties, needs, and challenges, as well as define requirements to guide the development of the prototype. The first impressions of these individuals are described, after an initial user study on the industrial shop floor, conducted to evaluate the proposed prototype, and collect feedback on how it can be improved to better support such contexts.

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1 INTRODUCTION

Industry scenarios are pushing the development of Augmented Reality (AR) solutions through the years [6, 14, 16, 18]. This technology is considered one of the nine pillars of Industry 4.0 given its ability to support operators for faster decision-making while improving work processes. This can be done by using real-time responsive computer-generated information superimposed over the real-world environment [3, 4, 7, 11, 17]. In fact, multiple prototypes have been proposed for different areas of application ranging from the assembly, inspection, quality control, maintenance, training, human-robot interaction, as well as co-located or remote collaboration, and others [1, 2, 10, 13, 15].

Nevertheless, the majority of studies have been limited to exploring a spatially limited area, which means having all the necessary materials within a limited range, otherwise, the tracking capabilities of existing tools could fail, affecting operators' task performance. To overcome this limitation, the concept of Pervasive AR offers experiences that can be visualized continuously as users move through the environment. Such experiences are also characterized by being aware of and responsive to the user's context and pose. However, few studies have combined Industry 4.0 real-time data monitoring and visualization with the advantages reported by this concept, in



Figure 2: Human-Centered Design (HCD) Methodology adopted: a) identify the needs of stakeholders on an assembly line; b) definition of requirements; c) design and development of a Pervasive AR-based prototype; d) interactive evaluation of the prototype.

particular, having applied AR prototypes in a real-life shop-floor context [5, 8, 9, 12].

This paper proposes a Pervasive AR prototype, created with partners from the industry sector, to support real-time data monitoring of an industrial assembly line. It also describes a set of first impressions collected based on the experience of nine individuals with distinct roles within the smart factory, where the preliminary tests occurred.

2 MONITORING DATA IN AN ASSEMBLY LINE USING AR

The prototype is the result of a Human-Centered Design (HCD) methodology, carried out through various remote and in-person meetings with industrial partners to identify stakeholders' difficulties, and challenges, as well as define requirements (Figure 2). The goal was to support line managers and data engineers with AR to analyze the status of the assembly line based on in-situ real-time data. The data is collected through various sensors spread across the assembly line and available through a REST API. In turn, this will enable to adapt distinct resources available to improve various *Key Performance Indicators* (KPIs). The requirements identified are:

- present current and past data of relevant KPIs;
- visualize an overview of each station status, as well as more detailed data on demand;
- identify bottlenecks and guide users to their position in space through directional means;
- provide hands-free interaction while visualizing superimposed data, allowing users to conduct physical tasks;
- update data every 5 seconds for retrieving real-time data measured by sensors installed on the assembly line.

Having the requirements identified, a Pervasive AR prototype, based on two modules (configuration and visualization), was developed to address them. It was created using the Unity 3D game engine and the Mixed Reality ToolKit (MRTK). The configuration module is used once, in the beginning, and is responsible for identifying each station location in the assembly line. For this, a virtual panel is placed close to the physical station and the station number is associated with the panel. The second module is responsible for visualization, where the data is organized and presented to the user. It accesses the configuration for positioning the visualization panels in the correct locations of the environment.

To elaborate, a set of features were designed and developed, as detailed next (Figures 1 and 3):

- **General visualization** - Each station in the line has its own visualization panel (Figure 1.1), which displays a summary of the information available (Figure 3.1) always facing the user.

These panels are divided into three general indicators (*cycle times*, *bottleneck*, and *KPIs*), each associated to a colored circle with different sizes representing the indicator state (green/small - all values are within the normal range; yellow/intermediate - a specific indicator is below 95% of the target value; red/big - a specific indicator is below 90% of the target value or the station is the bottleneck). This feature allows users to look around the assembly line and identify which stations have indicators outside of what is expected. Plus, the *general visualization panels* can be filtered to focus one of the indicators, hiding the remaining two indicators from the panel;

- **Specific visualization panel** - The data regarding a specific station can be observed in detail. The user can select one of the panels to obtain more information about it (Figure 3.1). This new panel will appear in front of the user, allowing them to navigate through the various indicators. The *Cycle Times* panel can be opened for presenting its specific indicators (*total cycle time*, *process time*, *exit time*, *waiting time*) (Figure 3.2). Each specific indicator is presented with the current value and with a colored circle (similar behavior to the *general visualization* regarding only the specific indicator). Identically, the *KPIs* panel shows its related indicators (*Overall Equipment Effectiveness* (OEE), *First-pass yield* (FPY), *PartCount*, *Productivity*);
- **Real time interactive charts** - The *specific visualization panels* can be further detailed through charts. By selecting a specific indicator, a line chart is drawn (Figure 3.3). The chart is composed of the last K indicator values measured. Using a slider, the time range (K) of the chart can be adjusted to a specific time interval;
- **Cloud-based integration** - To present the values in real-time and access the previous indicators values, the prototype is connected to a REST API. Every 5 seconds, the application updates its values, retrieving the required data to refresh the stations' information;
- **Bottleneck identification** - When the *general visualization panels* are *filtered* by the bottleneck indicator, the station displaying the line (if it exists) will be identified using a directional arrow pointing at it. This arrow can be followed through the environment to quickly reach the corresponding station (Figure 3.4). If the bottleneck station changes, a warning message will pop up, and the arrow will change its direction accordingly.

3 FIRST IMPRESSIONS ON THE SHOP FLOOR

To evaluate the Pervasive AR prototype, a three-hour visit to our industrial partner's assembly line was conducted. The goal was to evaluate the acceptance of different stakeholders towards the use

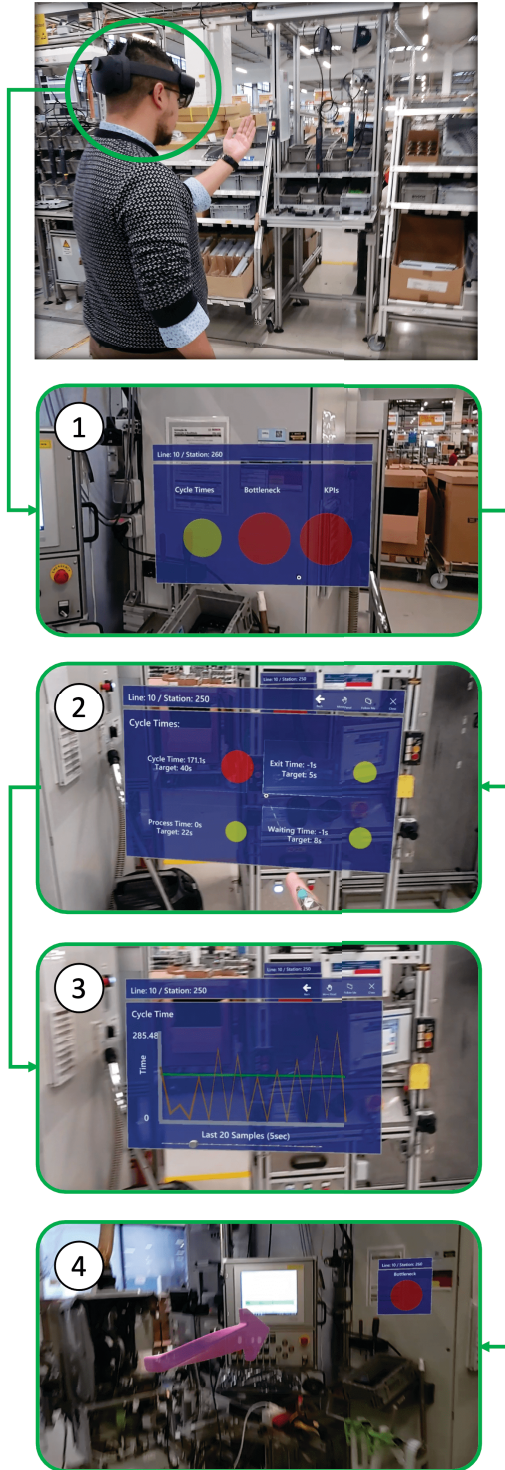


Figure 3: Pervasive AR visualizations for a given station of the assembly line: 1- dashboard illustrating different indicators to be explored; 2- detailed information for the cycle times (e.g., process, exit times, etc.); 3- chart with the last 20 samples of the total cycle time; 4- directional arrow pointing to the bottleneck station of the assembly line.

of Pervasive AR in a real-world shop floor scenario. Furthermore, collect first impressions regarding different topics of interest, namely usability, interaction, and tracking. Thus, establishing the next steps of our research.

In total, nine individuals (1 female, 8 male) from distinct departments of the factory participated in the study (e.g., line operator, line manager, production manager, process engineer, and quality engineer, among others). From these, 44.4% (4 out of 9) of participants had previous know-how or experience with AR, and only 22.2% (2 out of 9) were familiar with the tasks defined.

As for the tasks, participants were asked to use an HMD (Microsoft HoloLens 2) and visualize real-time data associated with an assembly line composed of various stations responsible for interactively assembling a final product. Moreover, identify certain aspects that could affect how to adjust the activities on the shop floor, e.g., identify the bottleneck and act accordingly based on the data displayed through the HMD. Also, editing the position of a given station, simulating the possibility of updating the arrangement of the assembly line stations in the future.

Regarding the procedure, participants were introduced to the concept of Pervasive AR, the goals of the study, as well as the experimental setup and design. After giving their informed consent, the defined tasks were completed, while being observed by a researcher who assisted them if necessary, and registered their actions and difficulties. At the end, a brief interview occurred, allowing participants to express their opinion about the prototype features, limitations, and possible improvements. The data collection process was conducted under the guidelines of the 1964 Helsinki Declaration.

Overall, participants considered the Pervasive AR prototype easy to use, having great potential to visualize in-situ real-time data of a single station or a plethora of visualizations regarding multiple areas of interest all at once. In fact, this was emphasized by all participants, showing that even users without expertise in AR technologies can easily obtain an overview of their surroundings, allowing them to better adjust operators' behavior based on the data collected. For example, the possibility to have a directional arrow to the assembly line bottleneck contributes to identifying where problems are occurring, to avoid the assembly line to become out of balance. In this vein, it is possible to more quickly intervene and allocate human resources (e.g., the less busy operator can help an overloaded operator).

Additionally, navigating through the different visualizations of a given station (from the main menu to a specific chart) was considered intuitive by the majority of participants, reporting that the information was easy to read and analyze. The use of charts was considered a nice add-on, particularly since the time interval can be adjusted to have a smaller or longer understanding of the assembly line indicators over time.

Besides, since all participants had different roles in the factory, this also shows that the prototype can be used by individuals with different backgrounds and that by using Pervasive AR, inexperienced users are also able to quickly accomplish more challenging tasks. This seems to illustrate that the prototype also has positive characteristics for training tasks moving forward, e.g., introducing a new operator to the selected tasks.

Another relevant insight mentioned by the participants was the fact that the shop floor environment can be easily updated, i.e., if a given station is reallocated, removed, replaced, or even if a new one is added, existing visualizations can be quickly selected and positioned in a different location in the assembly line. Some even tried to grab visualizations far from their position to avoid moving at all.

Regarding tracking, participants emphasized that moving through the space in a pervasive manner, without the need of markers, beacons, or similar, was relevant, providing more freedom to explore the space. As for interaction capabilities, participants needed an initial adaptation period but were able to quickly handle the learning curve,

even highlighting that with training, this could be easily overcome by any type of user.

One last remark, some participants were curious about the prototype's capacity to adapt to other assembly lines, or even if this technology could be introduced in distinct areas of application like logistics or picking processes, tasks which also took place in the factory. Likewise, there was also some interest regarding the device used, in particular, its battery lifetime, its weight, if it can be used during a full 8-hour shift, and also if it supported other interaction modalities, namely voice recognition or integration with an additional tablet for example.

As expected, participants also suggested a set of improvements, deemed relevant to refine the Pervasive AR prototype for the future. To elaborate, the following were mentioned:

- incorporate a warning message for situations when a given station does not have data available in the API;
- add a bottleneck shortcut in the initial menu, to speed up selecting this feature;
- integrate two indicators to the KPIs menu, which were previously not considered, but are now available in the API;
- enable the selection of visualizations from distant stations to avoid excessive movement in the assembly line;
- consider an overview menu showing every station closer to the user's position;
- allow seeing specific values in the charts used to increase the capacity to analyze a given situation.

4 FINAL REMARKS AND FUTURE WORK

First impressions from different stakeholders (e.g., line operators, line managers, process development engineers, and others) have been collected while assessing the potential of a Pervasive AR prototype for real-time data monitoring of an industrial assembly line. Results suggest that the proposed solution is viewed as extremely important for the future of the shop floor, as means to visualize synchronous data associated with each pre-defined station, as well as have a grasp of the assembly line as a whole. Furthermore, helping to identify bottlenecks that must be quickly fixed to ensure the levels of productivity are achieved as expected, was also emphasized as a game-changer. In fact, some participants mentioned that in the future, the prototype could also integrate the possibility to present suggestions regarding how to react to the bottlenecks (i.e., adjust other variables of the assembly line to compensate) based on all data aggregated.

This work is being expanded by conducting a larger user study with individuals from different shifts, assembly lines, and fields within the partner factory, allowing to understand how they react to the use of Pervasive AR technology during longer periods of time. Hence, helping to address how human factors (physical and cognitive workload, ergonomic considerations, fatigue, etc.) must also be incorporated to improve the prototype capacity to be adopted and used over larger amounts of time and situations. In addition, continue to develop relevant visualizations providing the state of the assembly line, as well as combine this with artificial intelligence and machine learning methodologies for introducing predictive maintenance features into the shop floor.

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